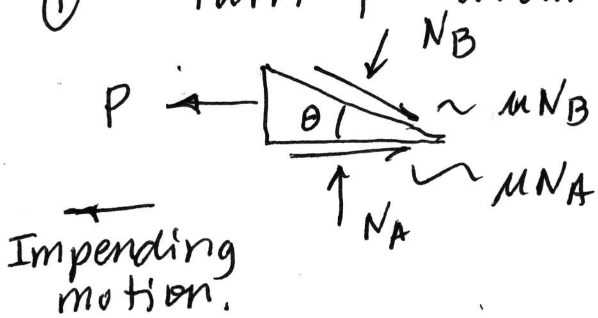
How to check self locking for a wedge?

In every case, do these two things:

① Turn P around (lowering case):



② Draw μN_A + μN_B terms opposing motion.

Once these are done, there are two ways to check for self-locking:

① Regardless of whether you know N_A or N_B, use the given μ to calculate the self locking θ.

$$\boxed{\theta_{SL} = \tan^{-1} \frac{2\mu}{(1-\mu^2)}}$$

If $\theta_{SL} > \theta_{given}$, then the wedge is self locking.

If $\theta_{SL} < \theta_{given}$, then the wedge will pop out (it is NOT self locking)

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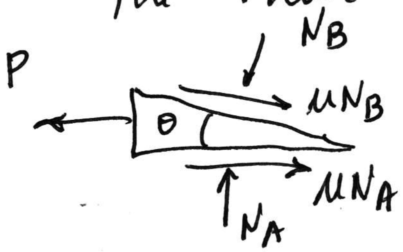
② If you know or can dermine N_B and N_A, use these plus the given θ and μ to calc the value of P.



If p is +, then a force is needed to pull the wedge out, thus the wedge is self locking.

If p is neg, then the wedge is NOT self locking.